Interview findings

1. How teams are currently formed

Teams are currently formed in several ways, and the method changes from module to module:

Mixed by course: Each team has one student from each course, for example one Computer Science student and one Software Engineering student.
Grouped by course: Teams are made only of students from the same course, for example a team of only Software Engineering students.
Balanced by performance: Students are spread across teams by their grades in earlier modules, so each team has a balanced mix of ability.
By attendance: Students are either balanced across teams by attendance, or grouped with students whose attendance is similar.
Random: Students are placed into teams at random.

Requirement: The system will provide two team-building algorithms:

Random allocation: students are placed into teams at random.
Balanced allocation: teams are balanced by student performance, weighted by attendance, for skill-based group work.

2. Why random allocation is common

Teams are usually allocated at random because teaching staff have little time to do anything else. Staff would prefer to form teams using set parameters, such as performance and attendance, so that teams achieve the best possible results.

3. Complaints and changing teams

Students often complain about their teams. It is often unclear whether a problem is caused by the random allocation itself or by how the team works together. More thought put into how teams are formed would likely lead to fewer complaints.

The interviewee raised the following points about how team changes should be handled:

Transparency. How teams are formed should be fully transparent to students and staff.
Staff options. Teaching staff need a range of options for how teams are formed.
Changes within the rules. Students should be able to change teams within reason, but changes must not break the rules a team was formed under. For example, a change cannot remove the only student from a required course.
Staff approval. All teams should be visible, and teaching staff must approve every change.

Requirement: The system will include a simple complaint system and a way for teaching staff to modify teams by hand.

4. Self-evaluation

Students evaluate their own performance. Teaching staff can then check these self-evaluations against grades, group results and other measures. Self-evaluation is useful, but the information needs to be monitored, because students may not judge their own performance accurately. More measures can be added to the system later if needed.

5. Attendance as the key measure

Attendance is the most important measure. Attendance and performance are almost always linked, and students who attend regularly nearly always do better.

Requirement: The system will record each student's attendance at meetings, lectures and labs.

6. Optional extras

Show a one-line reason for every student's team, for example "placed here to balance attendance." This delivers the full transparency they asked for in point 3, and it gives staff an answer when a student complains.

For each team, show its average grade and attendance next to the cohort average. Staff can then see at a glance that the balanced method worked, rather than taking it on trust.